(No) effects of shadowing on predictive eye movements during non-dominant language comprehension

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Some theories of prediction posit that people primarily use their production system to predict upcoming language during comprehension [1,2]. According to this prediction-by-production account, listeners covertly imitate the ongoing speech and run their production commands to pre-activate possible continuations of the sentence. However, the production-based mechanism is optional and may not be engaged when time or resources are limited, such as during non-dominant language comprehension [3]. Under such circumstances, a relatively automatic mechanism (e.g., pre-activation based on lexical priming) may account for prediction. Against this background, we investigated whether prediction in a non-dominant language can be enhanced by shadowing. We chose the shadowing task because it involves imitating ongoing speech assumed in the prediction-by-production mechanism and overt production that should engage the production system to a greater extent. Based on prior work showing a positive relationship between prediction and production [4,5], we hypothesised that shadowing may enhance prediction in non-dominant language by engaging the production system used for prediction.

English-dominant English-Chinese early bilinguals (N = 80) merely listened to (listening task) or listened to and shadowed (shadowing task) 28 Chinese sentences while viewing four objects (**Fig. 1**). We manipulated the main verb so that the target object was either predictable or unpredictable based on verb constraints [6]. We expected that participants would be more likely to look at the target object before it was mentioned in the predictable condition than in the unpredictable condition. Crucially, if the production system is engaged in prediction, we expected these predictive eye movements to be greater when participants shadowed than when they merely listened, because the shadowing task engages the production system. However, the shadowing task is likely to increase processing demands due to the additional production task, and this may interfere with other processes like prediction.

We ran cluster-based permutation tests in the time window from -1100 ms to 200 ms relative to the target word onset to test main effects of predictability and task, and their interaction. Participants were more likely to fixate the target in the predictable condition than in the unpredictable condition before the target word was mentioned, replicating [6]. However, this effect was similar in the listening task (cluster from -560 ms to 200 ms, cluster mass statistic = 2202, $p < .001$) and in the shadowing task (clusters from -640 to -600 ms and -560 ms to 200 ms, cluster mass statistic = 15 and 2113, $p = .02$ and $p < .001$). The main effect of task was not significant either.

The robust effect of predictability suggests that early bilinguals used verb constraints to predict the target during non-dominant language comprehension. However, we found no evidence that the shadowing task facilitated prediction. Shadowing may have no influence on prediction or comprehension. Alternatively, shadowing may have elicited two opposing effects that cancelled each other out: it may have facilitated prediction by engaging the production system more than mere listening, while simultaneously interfering with prediction due to its cognitive demands. It is also possible that the predictability effects we observed are ceiling effects and may not be further enhanced. Finally, the verb-based prediction we investigated may not necessarily involve the optional prediction-by-production system but may be entirely driven by an automatic priming-based mechanism (e.g., the verb *drink* pre-activates drinkable items via spreading activation).

Predictable 下课之后，小红要去喝那边的咖啡。
'After the class, Xiaohong is going to **drink** the **coffee** there.'

Unpredictable 出门之前，小红要去拿那边的咖啡。
'Before going out, Xiaohong is going to **take** the **coffee** there.'

Fig. 1. An example item. The sentence-final target (coffee) is predictable after the verb 'drink' but unpredictable after the verb 'take.'

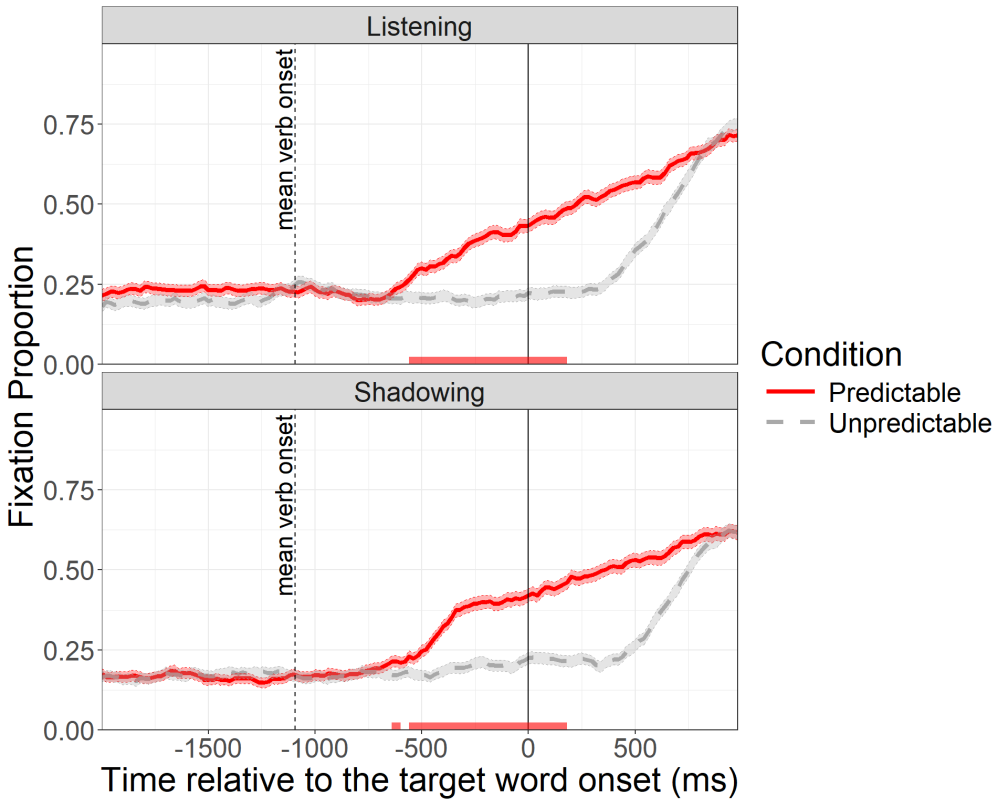

Fig. 2. Fixation proportion to the target. The red bars at y = 0 correspond to clusters that showed a significantly higher fixation proportion in the predictable vs. unpredictable condition in each task.

References

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